Here $\overset{\frown}{AB}$ is a minor arc in Fig.4.65, a semi circle in Fig.4.66 and a major arc in Fig.4.67. The point C makes different types of angles in different positions (Fig. 4.65 to 4.67). In all these circles, $\overset{\frown}{AXB}$ subtends $\angle AOB$ at the centre and $\angle ACB$ at a point on the circumference of the circle.

We want to prove $\angle AOB = 2\angle ACB$. For this purpose extend CO to D and join CD.

$\angle OCA = \angle OAC$ since $OA = OC$ (radii)

Exterior angle = sum of two interior opposite angles.

$$\begin{aligned}\angle AOD &= \angle OAC + \angle OCA \\ &= 2\angle OCA \quad \ldots. (1)\end{aligned}$$

Similarly,

$$\begin{aligned}\angle BOD &= \angle OBC + \angle OCB \\ &= 2\angle OCB \quad \ldots. (2)\end{aligned}$$

From (1) and (2),

$$\angle AOD + \angle BOD = 2(\angle OCA + \angle OCB)$$

Finally we reach our result $\angle AOB = 2\angle ACB$.

From this we get the result as follows :

Progress Check

i. Draw the outline of different size of bangles and try to find out the centre of each using set square.

ii. Trace the given cresent and complete as full moon using ruler and compass.



Fig. 4.68

Theorem 10

The angle subtended by an arc of the circle at the centre is double the angle subtended by it at any point on the remaining part of the circle.

Note

- Angle inscribed in a semicircle is a right angle.
- Equal arcs of a circle subtend equal angles at the centre.

Example 4.7

Find the value of $x°$ in the following figures:

(i)

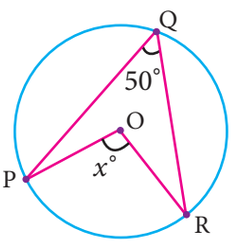


Fig. 4.69

(ii)

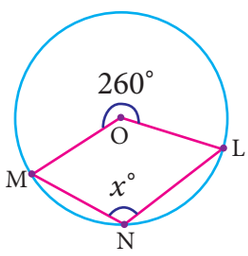


Fig. 4.70

(iii)

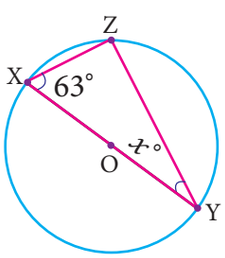


Fig. 4.71

(iv)

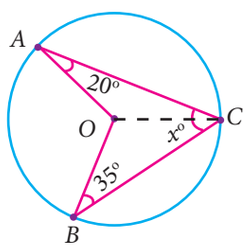


Fig. 4.72

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